

Alberto Progida

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Personal Profile

Spacecraft engineer with hands-on experience across the full hardware stack (mechanisms, composite structures, avionics, harnessing, and AIT) comfortable owning problems end-to-end in fast-moving teams. Previously, a first engineer at a two-founder space robotics startup, where I built a 3-DOF demonstrator in under two months that helped raise a £70k pre-seed round and secure the team's first letters of intent. Most recently at Oxford Space Systems, I led mechanism design, environmental qualification campaigns, and PDR reviews for a Ka-band antenna programme. I thrive working across disciplines and in small, committed teams: taking ownership of consequential decisions and executing with technical rigour and clear communication.

Engineering Experience

OSS - Oxford Space Systems Ltd.

Oxford, England, UK

Associate Spacecraft Mechanical Engineer

Sept 2024 - Current

- Owned the design and kinematic modelling of a **Ka-band, hinged rib antenna** deployment motorization chain to **ESA ECSS standards**,
- Performed detailed design of metallic mechanisms and composite structures,
- Planned and executed **DM-level testing**, implementing design modifications to reduce assembly time and engaging with suppliers to reduce procurement costs,
- Secured buy-in from Chief Engineering and **funding agencies** by leading trade-offs and presenting technical design packages at **PDR-level reviews**,
- 6-months clean-room based AIT activities for a high-stakes, multi-million pound **test campaign of EQM hardware** from piece parts to qualified system,
- Specification and execution of TVAC/Environmental/Vibration tests both internally and externally.

Lodestar Ltd.

London, England

Robotics Engineer

July 2023 - Sept 2023

- Worked closely with two founders in the search for product-market fit by compiling reports, carrying out market research and brainstorming,
- Designed, Built, Tested a **scaled 3-DOF demonstrator** in <2 months which was used to raise an initial £70k pre-seed round from Entrepreneurs First (EF),
- Designed, sized and assembled a modular powertrain of BLDC motors and Harmonic Drives,
- Developed a parametric preliminary design baseline for a 7-DOF LEO-capable robotic arm detailed with power, thermal, packaging, and cost budgets used for early customer outreach and to **secure the team's first 2x LOIs**.

ORBEX - Orbital Launch Express Ltd.

Forres, Scotland

Structures Engineering Placement (with Return Offer)

Sept 2022 - July 2023

- Owned the DFM of metallic/composite flight hardware, including **tolerance stack-up and GD&T to BS8888**,
- Designed and analysed metallic pressure vessel components and **bolted & fluid connections operating at cryo.**,
- Liaised with machinists and assembly technicians, leading meetings to **collaborate on clear and safe AIT procedures**,
- Managed and curated **>3000-part production-quality CAD assemblies**.

Flowcopter Ltd.

Edinburgh, Scotland

Avionics Hardware R&D Intern

June 2022 - Sept 2022

- Hardware reverse engineering** of ROTAX light aircraft ECU (Engine Control Unit) for autonomous operation,
- Design, component selection and test of **multi-layer PCBs** for data conditioning and high-power switching up 500W,
- Hardware-level integration of Data Acquisition, Engine and Hydraulic Pump Controllers and Autopilot,
- Manual Machining** of avionics housings and brackets.
- Instrumentation and control **wire harness planning** & procurement

RFA - Rocket Factory Augsburg AG

Augsburg, Germany

Hardware Test Engineering Intern

June 2021 - Aug 2021

- Wiring, testing and **commissioning of instrumentation** on a major MW-scale turbo-machinery test rig,
- Conceptualisation, Design and operation of pneumatic test rig for acceptance testing of COTS valves,
- Read, created and interpreted P&IDs for hydrostatic testing and safe operation of high pressure systems.

Education

Imperial College London

London, UK

MSc Advanced Aerospace Engineering - **Distinction - 85.6% - Second of my cohort**

Sept 2023 - Sept 2024

- **Thesis:** FEA, Manufacture and Characterisation of a Composite Flexure with Integrated Electrical Connections
- **Relevant Courses:** Spacecraft Structures, Systems Engineering, Composite Structures, Advanced Propulsion

University of Edinburgh

Edinburgh, Scotland

BEng Mechanical Engineering (Hons) - **First Class - 77%**

Sept 2019 - Jul 2022

- **Dissertation:** Design of a Subsea Actuator for Sensing Applications in Energetic Tidal Channels — 86%
- Institution of Mechanical Engineers (IMechE) Best Student Award

United World College of the Atlantic

St. Donat's Castle, Wales

International Baccalaureate - **Full ride scholarship, 41/45, top 2%**

May 2017 - May 2019

Projects

Composite Tape Spring with Integrated Electrical Connections

London, England

Structural Meta-Materials Research Group - Imperial College London

Jan 2024 - Sept 2024

- MSc Thesis supervised by Dr. Matthew Santer, now published in the *CEAS Space Journal*
- Experimenting with **ultra-thin printed circuits** (<25um) for lamination within a high-strain CFRP flexure
- Development of novel **layup techniques** to investigate the impact of the circuits on flexure performance
- **Optical and Scanning Electron microscopy** to investigate layup quality and identify defects
- Built and successfully correlated an **Abaqus Explicit Dynamic model** with experimental deployments

Chief Engineer - Darwin III Sounding Rocket Program

Edinburgh, Scotland

endeavour Student Rocketry Team

Sept 2021 - Oct 2022

- **Led 40+ students** in the design, manufacturing and successful **launch of a supersonic sounding rocket**,
- Collaborated with subsystem leads to specify technical requirements achievable within our timelines and **£25k budget**.
- Successfully launched the 34kg vehicle to 7000m, reaching a **top speed of Mach 1.7**
- Used **Altium Designer to create and test 10+ PCBs** throughout the Avionics Bay and Payload Section
- Digital and Analog Electronic Design, Serial communications, RF component selection
- Hands-on manufacture of Carbon and Glass-Fiber Composite structures (**Filament Winding, Pre-Preg and Wet Layup**)

Skills & Interests

Mechanical

ECSS Standards (design/testing) · GD&T (BS8888) · 3D Printing · FEMAP Thermal · Abaqus (Explicit/Standard) · FEMAP · SolidWorks · Composite layup & processing · DFM · AIT operations · Manual Machining · Hand Calculations

Electrical

KiCad · Altium Designer · PCB design · Schematics · Harness design · High-power switching · Board Bring-up

Programming

Claude Code · Python (Pandas, NumPy, SciPy, Numerical Methods, GUIs, Data Visualisation) · Matlab · PowerBI, ~~Excel~~

Interests

Long distance running, sailing, hiking, hard sci-fi literature & fresh pasta

Achievements

2026 **Published Article**, Integrated electrical connections in deployable flexures ([link to paper](#))

London, UK

2022 **Best Student**, Institution of Mechanical Engineers (IMechE)

Edinburgh, UK

2022 **First Prize**, European Rocketry Challenge: Best Payload - Awarded by ESA

Ponte de Sor, PT

2022 **BrightSpark**, Promising Young Engineer Award ([link to RS Comp./Electronics Weekly](#))

IET London, UK

2021 **First Prize**, Mach 21: Best CanSAT

Machrihanish AFB, UK

2017 **4th Place Worldwide**, Global ZeroRobotics Competition

ESTEC Noordwijk, NL

Languages

Italian

Native proficiency

English

Bilingual proficiency

French

Conversational B1